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SUPERSEDING
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MILITARY SPECIFICATION

DOLLY, TRAILER CONVERTERS

This specification is approved for use by US Army Tank-Automotive Materiel Readiness Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers tactical dollies for use in converting various semitrailers to full trailers.

1.2 Classification. Dollies shall be of the following models (see 6.2):

Model M197A1	- Dolly, Trailer Converter, 6 Ton, 2 Wheel
Model M198A1	- Dolly, Trailer Converter, 8 Ton, 2 Wheel
Model M354	- Dolly, Trailer Converter, 15 Ton, 4 Wheel, W/E

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

VV-B-680	- Brake Fluids, Automotive.
VV-L-800	- Lubricating Oil, General Purpose, Preservative, (Water-Displacing, Low Temperature).
ZZ-I-550	- Inner Tube, Pneumatic Tire.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Materiel Readiness Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2330

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MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-L-3150	- Lubricating Oil, Preservative, Medium.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-T-12459	- Tire Pneumatic: For Military Ground Vehicles.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-V-62038	- Vehicles Wheeled: Preparation for Shipment and Storage of.
MIL-B-46176	- Brake Fluid, Silicone, Automotive All Weather, Operational and Preservative.

STANDARDS

FEDERAL

FED-STD-595	- Color.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-193	- Painting Procedures, Tactical Vehicles (Tracked- and Wheeled).
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.

DRAWINGS

ARMY

8736120	- Dolly, Trailer Converter, 6 Ton, 2 Wheel, M197A1.
8736124	- Dolly, Trailer Converter, 8 Ton, 2 Wheel, M198A1.
8736308	- Dolly, Trailer Converter, 15 Ton, 4 Wheel, M354.

SUPPLEMENTARY QUALITY ASSURANCE PROVISIONS

MASTER LIST

8736120	- Supplementary Quality Assurance Provisions for Dolly, Trailer Converter, 6 Ton, 2 Wheel, M197A1.

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8736124 - Supplementary Quality Assurance Provisions for Dolly, Trailer Converter, 8 Ton, 2 Wheel, M198A1.

8736308 - Supplementary Quality Assurance Provisions for Dolly, Trailer Converter, 15 Ton, 4 Wheel, M354.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article (initial production vehicle). The first production vehicle of each model to be delivered under the contract shall be examined by Government personnel to determine conformance to the requirements of this specification. Vehicles submitted by the contractor shall be production models which are representative of the units to be delivered under the contract.

3.1.1 Endurance. The initial production vehicle, when operated in accordance with table III, shall possess not less than .60 probability of completing a minimum of 6000 miles of operation without replacement or overhaul of the principal components and/or subassemblies, i.e., suspension, including springs, axles and wheels, major brake system components (less brake shoes and common hardware), landing device, lunette, drawbar and frame. The criteria for replacement or overhaul will be as prescribed by the maintenance manual. The 6000 miles of operation shall be apportioned as follows:

a.	Hard-surfaced roads	40 percent
b.	Munson improved gravel roads	25 percent
c.	Level cross-country (Perryman No. 1)	10 percent
d.	Hilly cross-country (Churchville "C")	10 percent

3.1.2 Maintainability. The total scheduled maintenance, excluding driver/crew checks and services, shall be between 4 and 8 manhours. The total unscheduled maintenance shall be between 0 and 12 manhours during the first 6000 miles of operation. This equates to a Maintenance Ratio (MR) of .013 to .067 at 20 miles per hour (mph) and a (MR) of .02 to .10 at 30 mph. The scheduled maintenance intervals shall be 3 months or 3000 miles of operation, whichever comes first. Maximum time to repair (only to include diagnosis, repair and verification) shall be as follows, using personnel normally employed:

a. Organization maintenance shall not exceed 3 manhours 95 percent of the time.

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b. Direct support maintenance shall not exceed 10 manhours 95 percent of the time.
c. General support maintenance - not required.

3.2 Materials. Materials specified herein and in referenced drawings, specifications, and standards shall be free of defects and imperfections that might affect serviceability and function of the finished product (see 6.6).

3.2.1 Ozone resistance. When rubber components are required to be ozone resistant, the degree of resistance shall be as specified in applicable specifications or drawings.

3.2.2 Qualified products. The contractor shall be responsible for using parts and assemblies from Qualified Products Lists (QPL's) whenever available. Contractor's inspection records shall list all QPL items by number and date of the QPL, name of the supplier and part or drawing number(s). When parts and assemblies are approved as qualified products, but not yet listed on the QPL, the contractor shall list the products by number and date of the approved document and name of the supplier(s).

3.3 Construction. Vehicle shall be constructed in accordance with drawings, specifications, and standards referenced in engineering drawings specified in table I, and as supplemented herein. If more than one unit of a particular component is used in or on the vehicle, the components used shall be identical in make, material, and quality. The riveting, welding practices, and quality shall be the same for each vehicle. All parts, subassemblies and assemblies requiring identification shall be identified in accordance with MIL-STD-130.

TABLE I. Engineering drawings.

8736120	- Dolly, Trailer Converter, 6 Ton, 2 Wheel, M197A1.
8736124	- Dolly, Trailer Converter, 8 Ton, 2 Wheel, M198A1.
8736308	- Dolly, Trailer Converter, 15 Ton, 4 Wheel, M354.

3.3.1 Seals. When fording or operating in mud, sand or snow, seals shall prevent entrance of foreign matter into the bearings. Bearing seals shall restrict the leaking of lubricants from bearings.

3.3.2 Electrical circuits. Electrical circuits shall maintain continuity from end to end, without evidence of internal or external shorts, during all vehicle operating conditions.

3.3.3 Controls. All electrical, mechanical, and hydraulic controls shall operate without malfunction under all vehicle operating conditions.

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3.3.4 Adjustment mechanisms. All adjustment mechanisms shall function properly, and maintain adjustment within required settings during all vehicle operating conditions.

3.3.5 Welding. Welding shall meet the quality requirements specified on the applicable drawing and Supplementary Quality Assurance Provisions (SQAP's).

3.3.5.1 Welding repairs. Welding repairs of any kind shall be made only when, and to the extent, specifically authorized by the procuring activity (see 6.2).

3.3.6 Tires and tubes.

3.3.6.1 Tires. Tires shall be class CC, style A, of MIL-T-12459.

3.3.6.2 Inner tubes. Inner tubes shall conform to group II of ZZ-I-550, of proper size for the tire specified.

3.4 Performance. The vehicle shall meet the performance requirements of this specification when fully equipped, loaded in accordance with table III, serviced with products specified in table II, and towed by appropriate military towing vehicle.

3.4.1 Environmental. The vehicle shall operate as specified herein, under extreme conditions of weather, in ambient air temperatures ranging from minus 50° to plus 125°F.

3.4.2 Service products. Servicing and lubrication shall be performed using standard service products specified in table II.

TABLE II. Service products.

Product use	Ambient air temperature	
	-65° to 0°F	-10° to + 132° F
Oil:		
For hydraulic brakes	MIL-B-46176	VV-B-680
For general purpose lubrication,	VV-L-800	MIL-L-3150
For shock absorbers	MIL-H-5606	MIL-H-5606
Grease:		
For sealed bearings	MIL-G-23827	MIL-G-23827
For general chassis lubrication, including wheel bearings	MIL-G-10924	MIL-G-10924

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3.4.3 Loads, towing speeds and trailing ability.

3.4.3.1 Highway operation. When loaded and towed at speeds specified in table III, in a straight line, on a relatively level, smooth paved surface, the path of the towed vehicle shall not deviate more than 3 inches to either side of the path of the towing vehicle.

3.4.3.2 Cross-country operation. Vehicle shall be towable at specified load and speed (see table III) over unimproved roads, or gently rolling terrain. When so operated in a straight line, vehicle shall track the towing vehicle without damage to itself or the towing vehicle. Stability shall be maintained.

TABLE III. Vehicle loads and speeds.

Model	Total combined weight 1/		Towing vehicle	Semi-trailer class	Speed	
	highway	cross-country			high-way	cross-country
M197A1	14,880	14,880	2 1/2 ton	6 ton	50	10
M198A1	19,450	19,450	5 ton	12 ton	50	10
M354	36,100	36,100	10 ton	25 ton	35	10

1/ Total combined weight of dolly and payload.

NOTE: If dolly and empty semitrailer reach the total combined weight requirement, no semitrailer payload is required.

3.4.4 Gradeability. When towed over a dry, hard-surfaced road, free of loose material, each vehicle shall exhibit stability and gradeability specified.

3.4.4.1 Longitudinal inclines. Each vehicle shall follow the towing vehicle, without weaving to an extent which adversely affects the controllability of the vehicle combination, when ascending or descending longitudinal inclines having a 15 percent grade.

3.4.4.2 Side slopes. Each vehicle shall follow the towing vehicle without slipping or upsetting when operating, either left or right, on 20 percent side slopes.

3.4.5 Turning ability. When coupled to the towing vehicle, operating in its minimum turning circle, each vehicle shall follow without cramping or side-slipping and without damage to, or interference between, the towed vehicle and the towing vehicle.

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3.4.6 Braking ability.

3.4.6.1 Service brakes. The combined service brakes of the vehicle and towing vehicle shall control, decelerate, and stop the vehicle combination within 30 feet from a speed of 20 mph on dry, hard, level, smooth surfaced roads.

3.4.6.2 Automatic actuation. The automatic breakaway device shall apply and maintain application of the vehicle brakes for not less than 15 minutes, upon breakaway from the towing vehicle.

3.4.7 Fording. Each vehicle shall ford hard-bottomed, salt or fresh water crossing of depths great enough to completely submerge the chassis to top of fifth wheel and shall withstand such submergence for periods up to 15 minutes duration without damage to the vehicle or its components.

3.4.8 Landing leg. Unless otherwise specified (see 6.2), vehicle shall have a landing leg. The landing leg shall exhibit capacity and adjustment to raise, support and hold its portion of the fully loaded vehicle and shall withstand, without damage, the strains imposed when coupling or uncoupling the towing vehicle. The landing leg shall extend and retract; and when retracted, shall provide ground clearance for cross-country operation.

3.5 Painting, marking and identification plates.

3.5.1 Painting. Painting shall be in accordance with MIL-STD-193. Colors shall conform to FED-STD-595.

3.5.2 Marking. Marking of vehicle shall be in accordance with drawings and MIL-STD-642.

3.5.3 Identification plates. Identification plates shall meet the requirements of MIL-P-514.

3.6 Safety. All exposed parts which are energized electrically shall be located, insulated, fully enclosed, or guarded so as to prevent hazards to operating or maintenance personnel, or interference with equipment functioning. All moving parts which are of such nature, or so located, as to be a hazard to operating or maintenance personnel shall be enclosed or guarded. Protective devices shall not impair operating functions.

3.7 Servicing. Vehicle shall be adjusted for immediate operation and shall be serviced for climatic conditions at the delivery point. Servicing shall be compatible with processing for shipment as specified by the procuring activity (see 5.1).

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3.8 Workmanship. Workmanship shall be of such quality as to assure that the vehicle and its components are free of defects that compromise, limit or reduce the capability of the vehicle in the performance of its intended use. In addition to general appearance, the defects listed in tables VII and VIII shall be considered to be cause for failure.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of the inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality assurance system. The contractor shall provide and maintain an effective inspection and quality assurance system, acceptable to the Government, covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed, provided that an equivalent control is included in the approved quality assurance procedure. The contractor shall notify the Government of, and obtain approval for, any change to the submitted procedure that might affect the degree of control required by this specification, or other applicable documents referenced herein.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods and procedures of the written inspection plan are being properly applied and (b) Government inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed upon procedures or instances of practices which might have an adverse effect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct defects discovered shall be cause for suspension of acceptance until corrective action has been taken, or until conformance of product to specified criteria has been demonstrated.

4.1.3 Qualified and service products. The contractor's records shall be reviewed to assure conformance to 3.2.2 and 3.4.2.

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4.1.4 Parts and components. Parts, components and assemblies shall be inspected in accordance with engineering drawings (see table I) and master list of Supplementary Quality Assurance Provisions (SQAP's) (see table IV).

TABLE IV. Supplementary Quality Assurance Provisions.

8736120	- Supplementary Quality Assurance Provisions for Dolly, Trailer Converter, 6 Ton, 2 Wheel, M197A1.
8736124	- Supplementary Quality Assurance Provisions for Dolly, Trailer Converter, 8 Ton, 2 Wheel, M198A1.
8736308	- Supplementary Quality Assurance Provisions for Dolly, Trailer Converter, 15 Ton, 4 Wheel, M354.

4.2 Classification of inspections. Classification of inspections shall be as follows:

a. First production vehicle inspection (see 4.3).
b. Quality conformance inspection (see 4.6).

4.3 First production vehicle inspection.

4.3.1 In-process inspection. During fabrication of the first production vehicle (see 3.1) an in-process inspection will be conducted by the Government to evaluate conformance of materials and workmanship to specified requirements. The dolly, suspension and electrical components shall be available for inspection. Inspection shall be made at the contractor or subcontractor's facility prior to application of primer and paint. Processing and welding procedures, quality system and inspection records will be reviewed and evaluated during this inspection.

4.3.2 Completed first production vehicle inspection. The first production vehicle of each model shall be road tested and completely inspected by the contractor to determine conformance to the requirements specified herein. The applicable road test shall be in accordance with 4.7.2.2. Upon completion, the contractor shall submit the vehicle (and make available all inspection records and certifications) to the responsible Government inspection element at contractor's plant for preliminary inspection.

4.3.2.1 Preliminary inspection. The vehicle shall receive a preliminary inspection by the government representative. The preliminary inspection shall be complete, except for road test requirements specified in 4.7.2.2, which will be conducted jointly during provisional inspection. Preliminary inspection shall include verification of contractor's correction of all deficiencies reported on in-process inspection specified in 4.3.1.

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4.3.2.2 Provisional inspection. The first completed production vehicle shall be subjected to provisional inspection at the contractor's plant by government representatives of TARCOM/TARADCOM, Product Assurance Directorate. At the time of this inspection, the contractor shall make available his inspection plan, inspection records and certifications pertinent to the vehicle and components. Contractor shall further provide any required inspection assistance.

4.3.2.3 Repair of defects. Defects found as a result of the foregoing inspections shall be corrected by the contractor at no cost to the Government. Failure of the contractor to promptly correct defects shall be cause for suspension of acceptance of vehicles until corrective action has been accomplished.

4.3.2.4 Vehicle disposition. After completion of inspection, the first production vehicle of a specific model shall remain at the manufacturing facility as a production sample, and be the last vehicle shipped on the contract. The vehicle may be released sooner at the discretion of the Government. The contractor shall service and maintain the vehicle during this period.

4.3.3 Final approval and acceptance. Final approval and acceptance by the Government of the first production vehicle of a specific model shall be withheld until the 6000 mile test, as specified in 4.4 has been completed and a final determination has been made regarding conformity of the vehicle to requirements. This shall include, but not be limited to, workmanship and materials.

4.4 First article (initial production) test. To determine conformance to section 3 (inclusive), after completion of the first vehicle inspection, two additional vehicles shall be selected from the first month's production, or from the first 20 produced, and subjected to examination (see 4.6.2) and all tests specified in table VI. Subsequent to examination and tests, the selected vehicles shall be subjected to the 6000 mile test specified in table V, at a site approved by the Government, which will require a maximum of 90 days. These tests are to be performed by the Government. The contractor shall furnish repair parts, as required, to support the tests at no cost to the Government.

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TABLE V. 6000 mile test.

Course	Mileage and speeds	Semitrailer payload
Hard-surfaced, paved	2400 miles at varying speeds up to maximum highway speed	Highway
Hilly cross-country (Churchville "C")	600 miles at varying speeds up to maximum cross-country speed	Cross-country
Level cross-country	1440 miles at varying speeds up to maximum cross-country speed	Cross-country
Munson improved gravel roads	1500 miles at varying speeds up to maximum cross-country speed	Cross-country
Belgian block	60 miles at varying speeds up to maximum cross-country speed (20 miles of this mileage will be unloaded)	Cross-country

4.4.1 Endurance verification. To determine conformance to 3.1.1, the endurance requirement will be verified at a 50 percent confidence level during the 6000 mile test specified in 4.4.

4.4.2 Maintainability verification. To determine conformance to 3.1.2, maintainability requirements shall be verified during 6000 mile test specified in 4.4. Maintainability of the test vehicle shall meet specified scheduled and unscheduled maintenance man hours at required intervals.

4.4.3 Test failure. Failure of either vehicle to comply with any of the requirements specified, or any deficiency of a workmanship or materials nature found during, or as a result of, the 6000 mile test, will be cause for rejection of the vehicle. Further, the Government shall refuse to continue acceptance of production vehicles until evidence has been provided by the contractor that action has been taken to correct the deficiency. Any deficiency found during, or as a result of, 6000 mile test shall be evidence that all vehicles accepted prior to completion of the 6000 mile test are similarly deficient, unless evidence satisfactory to the contracting officer is furnished that they are not similarly deficient. Such deficiencies on all vehicles shall be corrected by the contractor at no cost to the Government.

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TABLE VI. Classification and location of inspection and tests.

Title	Requirement	First article 1/ test (4.4)	Acceptance 2/ test (4.7.1)	Control test (4.7.1)	Comparison 1/ test (4.7.3)
First article (Init. Prod)	3.1	4.4			
Endurance verification	3.1.1	4.4.1			
Maintainability verification	3.1.2	4.4.2			
Seals examination	3.3.1	4.5.1			4.5.1
Electric circuit examination	3.3.2	4.5.2	4.5.2	4.5.2	4.5.2
Controls examination	3.3.3	4.5.3	4.5.3	4.5.3	4.5.3
Adjustment mechanism examination	3.3.4	4.5.4	4.5.4	4.5.4	4.5.4
Welding repair examination	3.3.5	4.5.5	4.5.5		
Environmental test	3.4.1	4.5.6			
Highway operation test	3.4.3.1	4.5.7	4.5.7	4.5.7	4.5.7
Limited cross-country test	3.4.3.2	4.5.8			4.5.8
Longitudinal grade test	3.4.4.1	4.5.9			4.5.9
Side slopes test	3.4.4.2	4.5.10			4.5.10
Turning ability test	3.4.5	4.5.11		4.5.11	4.5.11
Service brakes test	3.4.6.1	4.5.12	4.5.12	4.5.12	4.5.12
Automatic actuation test	3.4.6.2	4.5.13		4.5.13	4.5.13
Fording test	3.4.7	4.5.14			4.5.14
Landing leg test	3.4.8	4.5.15	4.5.15	4.5.15	4.5.15
Painting	3.5.1	4.5.16.1	4.5.16.1	4.5.16.1	4.5.16.1
Marking	3.5.2	4.5.16.2	4.5.16.2	4.5.16.2	4.5.16.2
Identification plates	3.5.3	4.5.16.3	4.5.16.3	4.5.16.3	4.5.16.3
Safety	3.6	4.5.17			4.5.17
Servicing and adjusting	3.7	4.5.18	4.5.18	4.5.18	4.5.18
Workmanship	3.8	4.5.19	4.5.19	4.5.19	4.5.19
Perservation, packaging and vehicle processing	Section 5	4.5.20	4.5.20	4.5.20	

1/ Government proving grounds
2/ Contractor's facility

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4.5 Conformance verification.

4.5.1 Seals examination. To determine conformance to 3.3.1, after the 6000 mile test and fording operation, all bearings shall be checked for dirt contamination and lubricant leakage.

4.5.2 Electrical circuit examination. To determine conformance to 3.3.2, current and voltage output levels and continuity of the electrical system shall be measured.

4.5.3 Controls examination. To determine conformance to 3.3.3, all controls shall be operated and examined for functional requirements.

4.5.4 Adjustment mechanism examination. To determine conformance to 3.3.4, all adjustment mechanisms shall be examined for required adjustment and function.

4.5.5 Welding, repairs and burning examination. To determine conformance to 3.3.5 and 3.3.5.1, all welding, welding repairs and burning operations shall be examined to determine that slag has been removed and approved procedures have been followed.

4.5.6 Environmental test. To determine conformance to 3.4.1, the vehicle, properly serviced and equipped, shall be operated at the temperatures specified. Vehicle shall evidence no damage as a result of such operation.

4.5.7 Highway operation test. To determine conformance to 3.4.3.1, the vehicle shall be coupled to the towing vehicle. The combination shall be operated on paved highway and shall meet the performance specified.

4.5.8 Cross-country operation test. To determine conformance to 3.4.3.2, the vehicle shall be operated over cross-country terrain, as specified, without damage to the vehicle or towing vehicle.

4.5.9 Longitudinal grade test. To determine conformance to 3.4.4.1, the vehicle shall be towed up and down longitudinal grades at cross-country speed and shall meet specified requirements.

4.5.10 Side slope test. To determine conformance to 3.4.4.2, the vehicle shall be towed on side slopes at cross-country speed and shall meet specified requirements.

4.5.11 Turning and tracking ability test. To determine conformance to 3.4.5, the towing vehicle shall be coupled to the vehicle, driven to the towing vehicle's minimum turning circle, to the right and to the left, without interference between the towing and the towed vehicles.

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4.5.12 Service brakes test. To determine conformance to 3.4.6.1, the vehicle combination shall be operated as specified and observed for stopping distance. Stopping distance shall not exceed that specified. During acceptance test (4.7.1), the vehicle braking system shall be operated to the extent necessary to insure proper operation and adjustment of service brakes on all wheels.

4.5.13 Automatic actuation brake test. To determine conformance to 3.4.6.2, the vehicles and towing vehicle shall be placed on 15 percent grade and the automatic break-away device actuated. The brakes shall hold the vehicle stationary for time specified, when uncoupled from towing vehicle.

4.5.14 Fording test. To determine conformance to 3.4.7, the vehicle shall be towed through water crossings at the depth and for the time period specified.

4.5.15 Landing leg test. To determine conformance to 3.4.8, the vehicle shall be coupled and uncoupled from the towing vehicle and the landing leg operated. The landing leg shall perform as specified.

4.5.16 Painting, marking and identification plates.

4.5.16.1 Painting. To determine conformance to 3.5.1, preparation and painting shall be examined during and after application.

4.5.16.2 Marking. To determine conformance to 3.5.2, vehicle markings shall be examined.

4.5.16.3 Identification plates. To determine conformance to 3.5.3, all identification plates shall be examined.

4.5.17 Safety evaluation. To determine conformance to 3.6, vehicle shall be examined to assure that hazards to operating and maintenance personnel have been eliminated or controlled. Equipment which creates hazards shall be considered to be failure.

4.5.18 Servicing and adjusting. To determine conformance to 3.7, servicing and adjustment of specified systems shall be examined.

4.5.19 Workmanship. Vehicles shall be examined to determine conformance to 3.8.

4.5.20 Preservation, packaging and vehicle processing inspection. Material and equipment shall be inspected, prior to shipment, to determine con-

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4.6 Quality conformance inspection.

4.6.1 Inspection provisions.

4.6.1.1 Lot size. An inspection lot shall consist of all vehicles of one type, from one day's production submitted at one time for acceptance inspection.

4.6.1.2 Sampling inspection. For the purpose of visual, dimensional, and primary functional inspection, a representative sample shall be selected from each inspection lot in accordance with inspection level II of MIL-STD-105. Before sampling, the contractor shall 100 percent inspect the first 20 vehicles to establish a process average (see 6.4), to allow normal sampling in accordance with MIL-STD-105.

4.6.2 Examinations. Visual, dimensional, and primary functional examination shall consist of examination of the vehicle for conformance to the applicable drawings and this specification. Examinations shall be performed against the classification of defects and with the Acceptable Quality Levels (AQL's) specified in tables VII and VIII. These examinations shall be performed during all phases of manufacturing and subsequent to road testing.

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TABLE VII. Major defects AQL - 25 defects/100 units.

Def. No.	Characteristics	Major	Method of Inspection
101	Frame	Structural, welding and riveting defects	Visual
102	Fifth wheel	Malfunction, improper assembly or installation, welding defects	Visual
103	Axle	Improper assembly or installation, welding defects	Visual & functional
104	Brake system	Malfunction, component damage, non-conformance	Visual
105	Electrical system	Malfunction, inoperative, damage	Visual & functional
106	Suspension system components, springs and shock absorbers	Damaged, leaks, noise	Visual
107	Wheels, tires and tubes	Tire or tube damage, not as specified	Visual
108	Safety chains, intervehicular cable and hose	Improper lengths, damaged, malfunction, missing	Visual & functional
109	Landing leg	Improper extension, retraction or stowage action, and locking action (raised or lowered)	Visual & functional
110	Lunette eye	Casting defects, size improper	Visual
111	Drawbar assembly	Raising or lowering action improper, welding defects	Visual

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TABLE VIII. Minor defects AQL - 150 defects/100 units.

Def. No.	Characteristics	Minor	Method of Inspection
201	Wheels, tires and tubes	Improper mounting, pressure, or stud torque	Visual & functional
202	Wiring and tubing	Improperly coded, protected, installed or assembled	Visual
203	Brake components	Improper assembly or installation, clearance inadequate	Visual
204	Electrical system components	Improper assembly or installation	Visual
205	Suspension system components; springs and shock absorbers	Improper assembly or installation	Visual
206	Drawbar assembly	Improper assembly or installation	Visual
207	Landing leg	Improper assembly or installation	Visual
208	Lunette eye	Improper assembly or installation	Visual
209	Tie downs, lashing hooks, and lifting devices	Missing, improperly located, welding defects	Visual
210	Reflectors	Missing, defective, improper installation	Visual
211	Paint	Application or color improper	Visual

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TABLE VIII. Minor defects AQL - 150 defects/100 units. - Continued

Def. No.	Characteristics	Minor	Method of inspection
212	Lube fittings	Defective, missing, improperly installed	Visual
213	Lubrication	Improper application	Visual
214	Marking, decals, data and instruction plates	Incomplete data, improper size or installation, missing	Visual
215	Protective coatings	Application, coverage and materials improper	Visual
216	Record forms	Missing, improperly processed	Visual
217	Tools	Missing, improperly processed	Visual

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4.6.2.1 Unclassified characteristics. All defects that have no effect on function, safety, interchangeability, or life, but are considered departures from good workmanship shall be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots, or ten lots or more within a thirty day period, will be added to the minor classification of defects with no increase in AQL's.

4.6.2.2 Recurring major deficiencies. A major deficiency (see 6.5) is recurring when the same defect occurs more than once in the same sample or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records, ("P") chart or approved equivalent, reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiency. The deficiencies shall be corrected when found.

4.6.2.3 Recurring minor deficiencies. A minor deficiency (see 6.5) is recurring if it occurs more than twice in the same sample or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiency. Correction of all deficiencies shall be accomplished prior to acceptance.

4.6.2.4 Examination failure. If a vehicle fails to pass any examination or tests specified herein, the government inspector shall stop acceptance until evidence has been provided by the contractor that corrective action has been taken.

4.7 Classification of tests.

a. Acceptance tests (see 4.7.1).
b. Control tests (see 4.7.2).
c. Comparison tests (see 4.7.3).

4.7.1 Acceptance tests. To determine conformance to section 3 (inclusive), each vehicle selected in accordance with 4.6.1.2 shall be examined as specified in 4.6.2 and operated for a distance of not less than 5 miles by the contractor. Vehicle shall be completely assembled and serviced. Acceptance tests shall be those specified in table VI.

4.7.1.1 Five mile road test. Vehicle, without payload, shall be operated over a five mile course of relatively smooth, hard-surfaced roads.

4.7.1.2 Test failure. If the vehicle fails to pass any acceptance test specified herein, the Government shall stop acceptance of subsequent vehicles until evidence has been provided by the contractor that corrective actions has been accomplished.

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4.7.2 Control tests.

4.7.2.1 Frequency. The Government shall select, at random, one production vehicle per month as a minimum, or one out of each 100 vehicles produced per month, but not more than two in any 30 day period. Contractor shall conduct a 50 mile road test on each vehicle selected.

4.7.2.2 Fifty mile test. Each selected vehicle, loaded with full or simulated payload, shall be operated for a distance of 50 miles, and subjected to all control tests specified in table VI. These tests shall be performed at the place of manufacture by the contractor. The test course shall be relatively smooth, approximately level, hard-surfaced roads.

4.7.2.2.1 Test failure. If the vehicle fails to pass any control test, the Government shall stop acceptance examination and testing on subsequent vehicles until conditions causing the failure have been remedied. Any defect found during, or as a result of, the test shall be prima facie evidence that vehicles accepted subsequent to the previously accepted control test vehicle are similarly defective, unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly defective. Such defects on all vehicles shall be corrected by the contractor at no cost to the Government. Another vehicle, with corrective actions implemented, shall subsequently be subjected to the control test.

4.7.3 Comparison test. The Government may select vehicles any time during the contract production period and subject them to all applicable tests listed in tables V and VI to reveal deficiencies of a workmanship or materials nature that may reduce effective operations in the field, and to compare existing quality with previous standards. These tests shall be conducted by the Government. Selection of vehicles shall be on a spot check basis. Comparison test vehicles shall be loaded as specified in table III. Vehicles selected shall not include any vehicle previously tested for conformance to 4.7.2. The Government may elect to conduct comparison tests of less then 6000 miles as specified in table V.

4.7.3.1 Test failure. Failure of any vehicle to comply with specified requirements, or any defects of a workmanship or materials nature, may be cause for refusal to continue acceptance of vehicles until objective evidence has been provided that corrective action has been taken. Any defects found during, or as a result of the test shall be prima facie evidence that vehicles are similarly defective, unless evidence is furnished by the contractor that they are not similarly defective. Such defects on all vehicles shall be corrected by the contractor at no cost to the Government.

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5. PACKAGING

5.1 Preservation/processing. Preservation/processing for shipment and storage shall be in accordance with MIL-V-62038 and the applicable Equipment Preservation Data Sheet for the level(s) specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. Vehicles covered by this specification are intended for use in converting semitrailers to full trailers for transporting military supplies and equipment, and for mounting special equipment, when required.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Model required (see 1.2).
c. Whether welding repair is authorized, and extent allowed (see 3.3.5.1).
d. Landing leg not required (see 3.4.8).
e. Selection of level(s) of preservation /processing for shipment and storage (see 5.1).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examination, tests and approval of the first article (see 3.1).

6.4 Process average. Sampling may be initiated if the process average value for the first twenty vehicles inspected is less than the AQL specified in the classification of defects for major and minor defects. Process average will be computed as follows:

$$\text{Process average} = \frac{\text{Number of defects}}{\text{Number of vehicles inspected}} \times 100$$

If the computed process average exceeds the specified AQL, 100 percent inspection should be performed and continued until such time that the process average for twenty consecutive vehicles is less than the specified AQL.

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6.5 Deficiency sheet. The government inspector will verify that a thorough inspection of each vehicle is performed by the contractor not only for the listed characteristics, but also for any departures from good workmanship. The government inspector will assure all deficiencies encountered during the inspection are enumerated on the Deficiency Sheet for the vehicle. The defects noted on Deficiency Sheet should contain sufficient description to enable the government inspector and the contractor's representative to classify the deficiency in accordance with the classification of defects (see tables VII and VIII) and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring deficiencies.

6.6 Recycled materials. The use of recycled materials which meet the requirements of the applicable materials specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

6.7 Changes from previous issue. Asterisks are not used in this revision, to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army - AT

Reviewer activity:
Navy - MC

User activity:
Navy - YD

Preparing activity:
Army - AT

Project No. 2330-0711

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-45322E(AT)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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